Dishan Sarkar

Kolkata, West Bengal
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Education

Birla Institute of Technology, Mesra **Sep 2023 – May 2027 (Expected)**
Bachelor of Technology (Artificial Intelligence And Machine Learning) CGPA - 8.5 *Ranchi, Jharkhand*

Skills

Languages: Python, Java, C++, SQL, JavaScript
CS Fundamentals: OOPS, DBMS, CN, DSA
AI/ML: PyTorch, Transformers, RAG, Multi-Agent Systems, LangGraph, FAISS, ChromaDB, Gemini
Backend: FastAPI, Spring Boot, Node.js, REST APIs, PostgreSQL, Redis, Kafka
Cloud & DevOps: AWS, Azure, Docker, Terraform, GitHub Actions, Linux, Git

Experience

Accenture **Kolkata, India**
Associate Software Engineering Intern *May 2026 – Jul 2026*

- Built an **AI-powered Incident Intelligence Platform** with ML-based ticket prioritization, duplicate detection, and automated triage for **500+ daily enterprise incidents**.
- Developed a **Natural Language-to-SQL analytics engine** and deployed containerized services on **Azure** (**Docker**, **PostgreSQL**), cutting reporting turnaround by **30%**.

Believerly Pvt Ltd **Remote**
Backend Developer *Jan 2026 – Mar 2026*

- Engineered a **Redis-backed token-bucket rate limiter** with per-IP and per-user quotas, sustaining **500+ requests/sec** while prioritizing critical transactions through asynchronous queues during traffic spikes.
- Built a **Kafka**-based SMS and email notification service with automated retries and real-time delivery-event tracking.

Projects

RecoveryAI | **AI Payment Recovery Platform** | *FastAPI, PostgreSQL, Gemini, Sarvam AI, Next.js* **GitHub**

- Built a **zero-trust** payment-recovery platform: Gemini only classifies intent, while a deterministic **policy engine** and **13-state FSM** decide every financial action, with **0 policy violations across 52 adversarial prompts**.
- Built **Hinglish/Hindi/English voice negotiation** (Sarvam STT/TTS) scoring **82% on held-out utterances**; made actions retry-safe with a **Postgres Idempotency-Key** and `SKIP LOCKED` scheduling (**14 property tests**).

LedgerFlow | **Double-Entry Payments Backend** | *Java, Spring Boot, PostgreSQL, AWS, Terraform* **GitHub**

- Built a UPI-style payments backend with a **double-entry ledger enforced in PostgreSQL** and **idempotency keys**: **0 paise imbalance across 520K+ journal lines** and **0 double charges** under concurrent retries.
- Sustained **175 payments/s at p99 under 200 ms with 0 errors** in k6 load tests; delivered **HMAC-signed webhooks** via a **transactional outbox** on SNS/SQS, deployed on AWS ECS and RDS with Terraform.

Agentic SOV Intelligence | **Insurance Data Cleansing** | *LangGraph, ChromaDB, Pandas, Pandera* **GitHub**

- Built a **four-agent LangGraph pipeline** that maps and cleanses chaotic insurance SOV workbooks into an exact 17-field schema: **55/55 real-file mappings** in **15–22 s** end-to-end.
- Designed a **cascading schema mapper** (memory, exact, fuzzy + semantic, then LLM) with **ChromaDB** feedback memory; the LLM never touches data, only **Pandera**-validated, human-approved, audited transforms run.

Research

Adaptive Financial Transformer with Regime-Gated Attention **BIT Mesra**
Independent Researcher | *arXiv* | *Zenodo DOI* *Jan 2026 – Present*

- Designed and evaluated Transformer-based models for 5-day stock return forecasting, with ablation studies, multi-seed benchmarking, and paired statistical significance testing.
- Reduced model parameters by **15.2%** and input features by **58%** at comparable predictive performance, validated with Directional Accuracy, Sharpe Ratio, and transaction-cost-aware backtesting.

Achievements & Certifications

Competitive Programming: Codeforces Specialist (max rating 1499) | 300+ LeetCode problems solved
Awards: Amazon ML Challenge Top 200 Teams | National Finalist, India Innovates 2026 | SIH Top-30 Institute Team | ISRO Bharatiya Antariksh Hackathon Shortlist

Certifications: CompTIA Security+, Network+, Cloud+ | Accenture GenAI & Agentic AI | DeepLearning.AI ML | Google Cloud Foundations

Leadership

Vice-President - IETE Student Chapter, BIT Mesra **August 2024 – Present**

- Directed technical initiatives and coordinated training programs for **220+ students**.